

DUTCH 4YOU

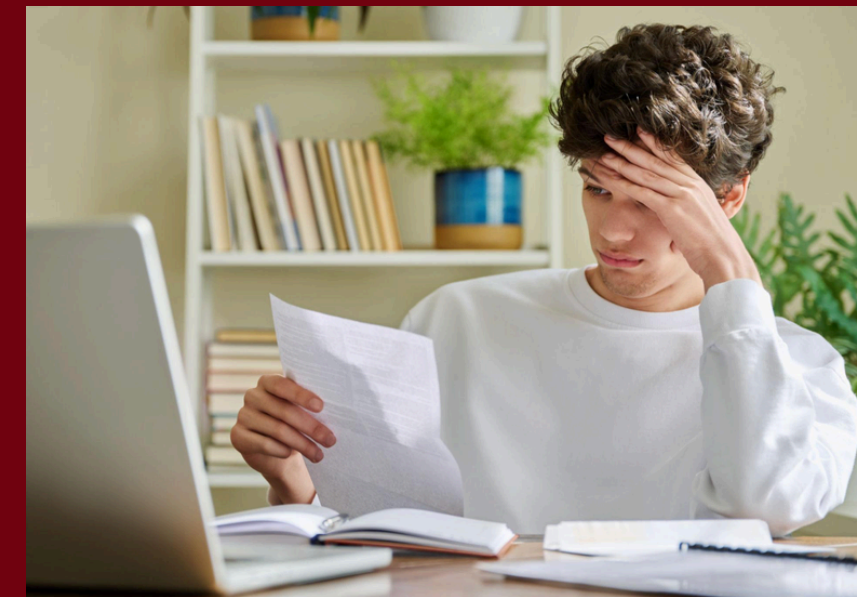


AI FOR GOOD — HACKATHON 3:
EQUAL ACCESS

PRESENTED BY :
Evaldas
Pavitra

**SDG 10 — REDUCED
INEQUALITIES**

PROBLEM



1.

IMPORTANT INFORMATION IS OFTEN WRITTEN IN DUTCH

Many important communications in the Netherlands are written in Dutch, making them difficult for international students who are still learning the language

2.

IMPORTANT INFORMATION GETS MISSED

Official messages can contain deadlines, payments, required documents, and actions that users need to understand quickly.

3.

UNEQUAL ACCESS TO INFORMATION

Students may receive the same information as everyone else but have more difficulty understanding and acting on it because of the language barrier.

WHO IS IT INTENDED FOR AND WHY DOES IT MATTER?

USER GROUP

International & exchange students in the Netherlands with limited Dutch proficiency

- Receive Dutch administrative messages
- Often need to understand them quickly and independently
- Typical messages: university, housing, municipality, bank
- Key information: deadlines, payments, documents, required actions

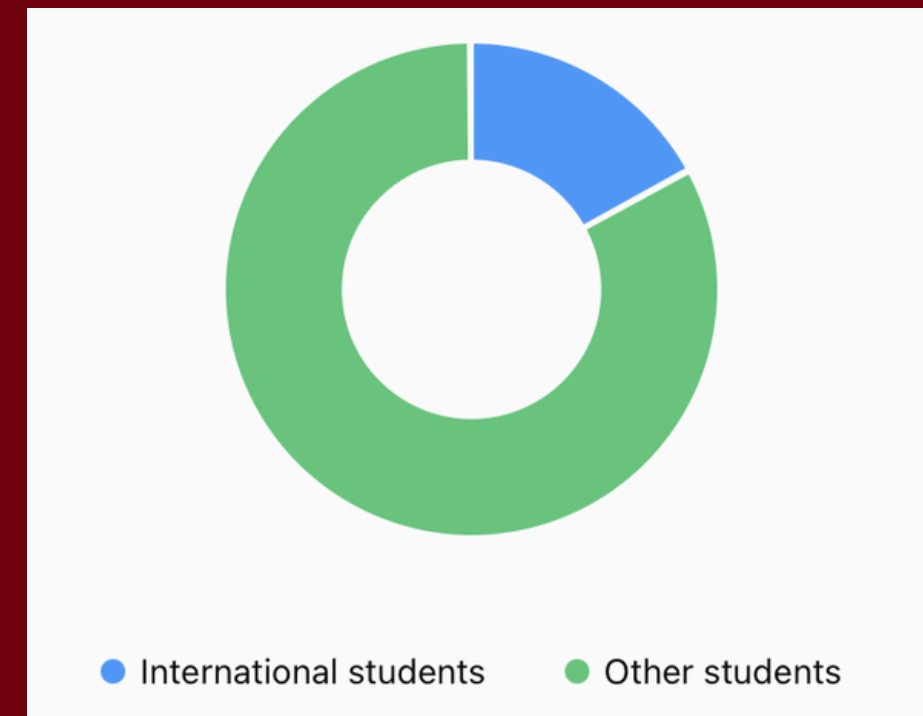
! WHY THIS USER GROUP MATTERS

A student may understand English well but not Dutch

The problem is not translating every word — it is identifying what the Dutch message requires them to do.

Missing a deadline, payment or document request can directly affect the student

Dutch4You turns the message into English + explanation + actionable information



EVIDENCE

129,764 international degree students
17% of all students in Dutch higher education
2025/26

Source: Nuffic, 2026

OUR SOLUTION

MEET DUTCH4YOU,

AN AI-POWERED TOOL THAT TURNS DUTCH COMMUNICATION INTO UNDERSTANDABLE, ACTIONABLE ENGLISH. WE DON'T JUST TRANSLATE THE MESSAGE. WE STRUCTURE THE INFORMATION THE USER NEEDS TO ACT ON IT.

User gives:
 Dutch letter / message



Dutch4You uses:

 Gemini API



User gets:

- English translation
- Simple explanation
- Important information
 - Deadlines
 - Amounts
 - Required actions
- Confidence & warnings

 Dutch4You

Got a letter in Dutch?

Paste it below. You get an English translation, a plain explanation, and a checklist of what to do and by when.

LIVE DEMO

[HTTPS://DUTCH4YOU.STREAMLIT.APP/](https://dutch4you.streamlit.app/)

Got a letter in Dutch?

Paste it below. You get an English translation, a plain explanation, and a checklist of what to do and by when.

Try an example

Housing

University

Municipality

Bank

U dient uiterlijk 30 september uw documenten in te leveren. Als u dit niet op tijd doet, kan uw aanvraag vertraging oplopen.

124 characters. Remove personal numbers such as your BSN or IBAN before pasting.

Translate and explain

Clear

PASTE THE IMPORTANT DOCUMENT OR DUTCH TEXT

In English

You must submit your documents by 30 September at the latest. If you do not do this on time, your application may be delayed.

Dutch original

U dient uiterlijk 30 september uw documenten in te leveren. Als u dit niet op tijd doet, kan uw aanvraag vertraging oplopen.

What this means

This notice informs you of a strict deadline to submit required documents. If you miss the September 30th deadline, the processing of your application will be delayed.

What you need to do

- ☒ Submit your documents

Deadline

30 September
(uiterlijk / at the latest)

Amount

No amount found

● High confidence

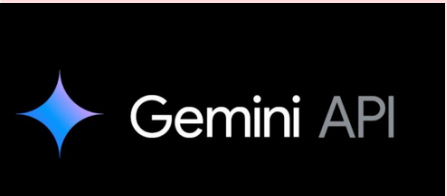
Save summary

EXPLAINS THE ENTIRE TRANSLATION,
GIVES YOU IMPORTANT DEADLINES AND DETAILS

TOOLS AND TECHNOLOGY



Component	Technology
Programming language	Python
User interface	Streamlit
AI model/API	Google Gemini API
Python AI SDK	Google GenAI SDK
Environment variables	python-dotenv
Deployment	Streamlit Community Cloud
Version control	Git + GitHub



SDG 10 - REDUCED INEQUALITIES



- **Target 10.2:** Promote the social, economic and political inclusion of everyone, regardless of factors such as origin or background.

HOW IS OUR PROJECT SUPPORTING SDG 10?

- **Limited Dutch → information barrier**
- **Makes deadlines, payments & actions easier to understand**
- **AI translates, simplifies & highlights key information**
- **Improves access to important information**
- **Dutch4you, Reduces a language barrier to inclusion**

ETHICAL CONSIDERATIONS



1.AI Accuracy — Gemini may produce an incorrect translation or explanation.

Safeguard: The system provides confidence information and warns users when the output may be uncertain.

2.User Dependence — Users with limited Dutch may not be able to verify the AI's interpretation themselves.

Safeguard: Users are advised to verify important information with an official source or Dutch speaker.

3.Unclear Input — The AI could misunderstand text that is unreadable or not Dutch.

Safeguard: The system is instructed not to guess and can return a low-confidence warning.

4.Technical Failure — API errors or invalid AI responses could produce unreliable results.

Safeguard: The application handles API failures, retries temporary errors, and rejects invalid JSON instead of treating it as reliable output.

5.Privacy — Official messages may contain personal or sensitive information.

Safeguard: Users are advised to avoid submitting highly sensitive information unnecessarily.

6.Scope of Use — The tool should not be treated as a certified translator or professional advisor.

Safeguard: Dutch4You clearly states its limitations and should not be relied upon for high-stakes legal or immigration decisions.

RELIABILITY & EDGE CASES:

HOW WE HANDLE UNRELIABLE INPUT & FAILURES.

- Empty input → asks the user to enter Dutch text
- Non-Dutch / unclear text → does not guess; shows a warning
- Low-confidence AI output → displays a confidence level and warning
- Invalid AI response → rejects invalid JSON instead of showing unreliable results

- API failure → handles errors and retries temporary failures
- Important information → separately extracts deadlines, amounts & required actions
- High-stakes information → advises users to verify with an official source or Dutch speaker
- Key point: If the system is uncertain, it communicates that uncertainty instead of presenting an answer as fact

● High confidence

Heads-up: The text mentions paying 'on time', but the specific due date and payment amount are not provided in this short snippet.

THANK YOU



THANK YOU